

EMCH 367 Controls

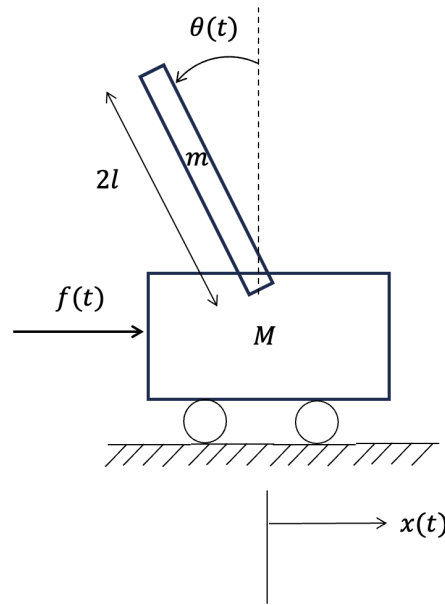
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Fall 2024

Homework #3

Due date: October 13, 11:59 pm

Problem 1. (60pt) Consider the following inverted pendulum system.



The equation of motion is given by

$$(M + m)\ddot{x}(t) + b\dot{x}(t) = f(t) - ml \sin \theta(t) \cdot \dot{\theta}^2(t) + ml \cos \theta(t) \cdot \ddot{\theta}(t), \quad (1)$$

$$(J + ml^2)\ddot{\theta}(t) - mgl \sin \theta(t) = ml\ddot{x}(t) \cos \theta(t), \quad (2)$$

where M is the mass of the cart, m is the mass of the rod, l is the half length of the rod, θ is the angle of the rod with respect to the upright position, b is the coefficient of friction for cart, μ is the friction coefficient between the cart and the ground, and J is the moment of inertia of the rod. Assume that friction force is modeled as $b\dot{x}(t)$.

(a) (10pt) Assume that we have small θ , that is, $\theta \ll 1$. Using the following simplification, re-write (1) and (2).

$$\sin \theta(t) \simeq \theta(t), \quad \cos \theta(t) \simeq 1, \quad \dot{\theta}^2(t) \simeq 0.$$

Solution: Using the simplifications, it follows that

$$\begin{aligned}(M + m)\ddot{x}(t) + b\dot{x}(t) &= f(t) + ml\ddot{\theta}(t), \\ (J + ml^2)\ddot{\theta}(t) - mgl\theta(t) &= ml\ddot{x}(t).\end{aligned}$$

(b) (10pt) Let $M = 1$ kg, $m = 0.5$ kg, $b = 0.2$ N/(m · s), $l = 0.4$ m, $J = 0.01$ k · m². Find the transfer function between $\Theta(s)$ as an output and $F(s)$ as an input, i.e., find $\Theta(s)/F(s)$. Assume zero initial condition.

Solution: Taking the Laplace transformation of the simplified dynamics yields

$$\begin{aligned} [(M + m)s^2 + bs]X(s) &= F(s) + mls^2\Theta(s), \\ [(J + ml^2)s^2 - mgl]\Theta(s) &= mls^2X(s). \end{aligned}$$

Now, it follows that

$$X(s) = \frac{(J + ml^2)s^2 - mgl}{mls^2}\Theta(s),$$

and hence, we have

$$[(M + m)s^2 + bs]\frac{(J + ml^2)s^2 - mgl}{mls^2}\Theta(s) - mls^2\Theta(s) = F(s).$$

Therefore, the transfer function is given by

$$\begin{aligned} \frac{\Theta(s)}{F(s)} &= \frac{mls^2}{[(M + m)s^2 + bs][(J + ml^2)s^2 - mgl] - m^2l^2s^4} \\ &= \frac{0.001805s}{0.0008574s^3 + 0.0001625s^2 - 0.02653s - 0.003538} \\ &= \frac{0.2s}{0.095s^3 + 0.018s^2 - 2.943s - 0.3924} \\ &= \frac{2.105s}{s^3 + 0.1895s^2 - 30.98s - 4.13}. \end{aligned}$$

(c) (10pt) Find the transfer function between $X(s)$ as an output and $F(s)$ as an input, i.e., find $X(s)/F(s)$. Assume zero initial condition.

Solution: Since

$$\begin{aligned} [(M + m)s^2 + bs]X(s) &= F(s) + mls^2\Theta(s), \\ [(J + ml^2)s^2 - mgl]\Theta(s) &= mls^2X(s), \end{aligned}$$

it follows that

$$\Theta(s) = \frac{mls^2}{(J + ml^2)s^2 - mgl}X(s),$$

and hence, we have

$$[(M + m)s^2 + bs]X(s) = F(s) + mls^2 \frac{mls^2}{(J + ml^2)s^2 - mgl}X(s).$$

Therefore, it follows that

$$\begin{aligned} \frac{X(s)}{F(s)} &= \left[(M + m)s^2 + bs - \frac{m^2l^2s^4}{(J + ml^2)s^2 - mgl} \right]^{-1} \\ &= \frac{(J + ml^2)s^2 - mgl}{[(M + m)s^2 + bs][(J + ml^2)s^2 - mgl] - m^2l^2s^4} \\ &= \frac{0.0008123s^2 - 0.01769}{0.0008574s^4 + 0.0001625s^3 - 0.02653s^2 - 0.003538s} \\ &= \frac{0.09s^2 - 1.962}{09499s^4 + 0.018s^3 - 2.943s^2 - 0.3924s} \end{aligned}$$

(d) (10pt) Using MATLAB, plot the impulse response of the transfer function found in (b) and (c) for 1 second.

Solution:

```

clc; clear all; close all;

M = 1;
m = 0.5;
b = 0.2;
I = 0.01;
g = 9.8;
l = 0.4;
s = tf('s');

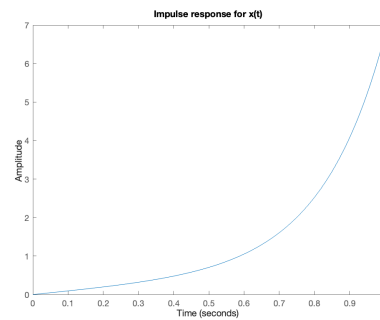
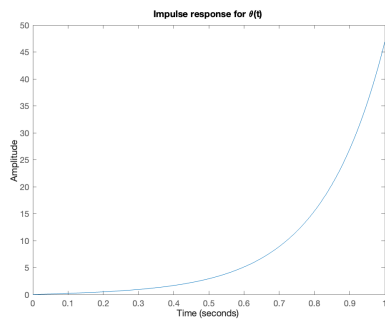
%% (b)
q = (M+m)*(I+m*l^2)-(m*l)^2;
G_theta = (m*l*s/q)/(s^3 + (b*(I + m*l^2))*s^2/q - ((M + m)*m*g*l)*s/q - b*m*g*l/q);

%% (c)
G_x = ((I+m*l^2)/q*s^2 - m*g*l/q)/(s^4 + (b*(I + m*l^2))*s^3/q - ((M + m)*m*g*l)*s^2/q - b*m*g*l*s/q);

%% (d)
figure(1)
impz(G_theta,1)
title('Impulse response for \theta(t)')

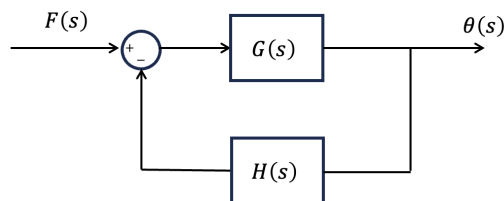
figure(2)
impz(G_x,1)
title('Impulse response for x(t)')

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(e) (10pt) Now, consider the following feedback system with $H(s)$ given by

$$H(s) = K_p + \frac{K_i}{s} + K_d s.$$



Plot three different impulse responses (10 seconds) for the closed loop system with the following three different gains. Note that $G(s) = \frac{\Theta(s)}{F(s)}$ from (b).

$$K_p = 50, \quad K_i = 0, \quad K_d = 1$$

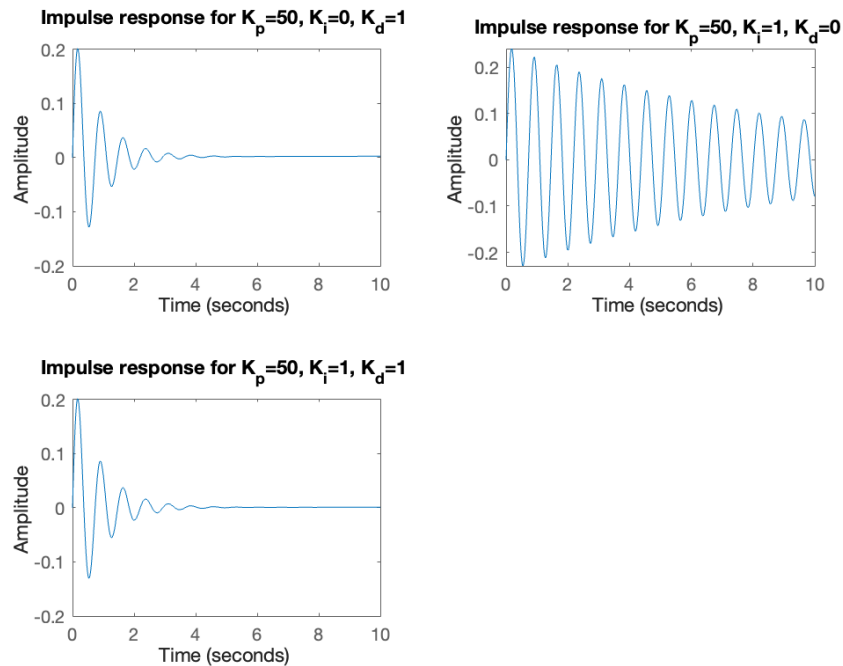
$$K_p = 50, \quad K_i = 1, \quad K_d = 0$$

$$K_p = 50, \quad K_i = 1, \quad K_d = 1$$

Solution:

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%% (e)
t=0:0.01:10;
Kp1 = 50; Ki1 = 0; Kd1 = 1;
Kp2 = 10; Ki2 = 2; Kd2 = 0;
Kp3 = 50; Ki3 = 1; Kd3 = 1;
H_1 = pid(Kp1,Ki1,Kd1);
H_2 = pid(Kp2,Ki2,Kd2);
H_3 = pid(Kp3,Ki3,Kd3);
G_cl1 = feedback(G_theta,H_1);
G_cl2 = feedback(G_theta,H_2);
G_cl3 = feedback(G_theta,H_3);

figure(3)
subplot(2,2,1)
impz(G_cl1,t)
title('Impulse response for K_p=50, K_i=0, K_d=1')
subplot(2,2,2)
impz(G_cl2,t)
title('Impulse response for K_p=50, K_i=1, K_d=0')
subplot(2,2,3)
impz(G_cl3,t)
title('Impulse response for K_p=50, K_i=1, K_d=1')
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(f) (10pt) Compare the results from (e). What are the differences among PD, PI, and PID controller? (Hint: Zoom in for the steady state.)

Solution: Comparing the first and second results reveal the difference between PI and PD control. As shown in the impulse responses, adding derivative control introduces damper to the system improving the convergence time. Additionally, integral controller is slower compared to the derivative control.

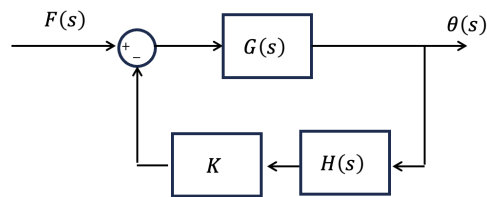
Comparing the second and third results reveal the difference between PI and PID control. Once you add derivative controller to PI control, it provides damping.

Comparing the first and third results shows the advantage of adding integral controller. If you look closely, PD controller has constant steady state error while PID controller reduces the constant steady state error to zero.

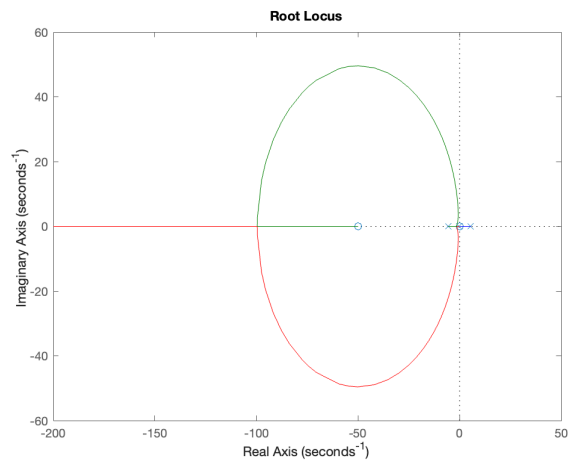
(g) (20pt) Consider the following feedback system with $H(s)$ given by

$$H(s) = 0.02s + 1 + \frac{0.02}{s}.$$

Draw the root locus with MATLAB and find the closed loop poles with $K = 50$.



Solution:



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%% (g)
H = 1*(1+0.02/s+0.02*s);
r = rlocus(G_theta*H,50);
figure(4)
rlocus(G_theta*H)
```

For $K = 50$, the corresponding roots are

$$-1.1610 + 8.5440j, \quad -1.1610 - 8.5440j, \quad 0.0272.$$

Problem 2. (20pt) Consider the following 2nd order system

$$\frac{X(s)}{F(s)} = G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

The step response has the following performance indicators

$$t_s = 4 \text{ s}, \quad t_p = 0.81 \text{ s}, \quad x_p = 1.44.$$

Find ζ and ω_n . (Round ζ to the nearest hundredth and ω_n to the nearest integer.)

Solution: Note that

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = 0.81,$$
$$x_p = 1 + e^{-\frac{\zeta}{\sqrt{1 - \zeta^2}} \pi} = 1.44,$$

and hence,

$$\frac{\zeta}{\sqrt{1 - \zeta^2}} \pi = -\ln(0.44) \implies \frac{\zeta^2}{1 - \zeta^2} = \left(\frac{\ln(0.44)}{\pi} \right)^2.$$

Therefore, it follows that

$$\zeta^2 = \frac{\left(\frac{\ln(0.44)}{\pi} \right)^2}{1 + \left(\frac{\ln(0.44)}{\pi} \right)^2} \implies \zeta = 0.2528 \simeq 0.25.$$

Next, note that

$$\frac{\pi}{\omega_n} = 0.81 \sqrt{1 - \zeta^2} \implies \omega_n = \frac{\pi}{0.81 \sqrt{1 - \zeta^2}} = 4.0057 \simeq 4 \text{ rad/s}.$$

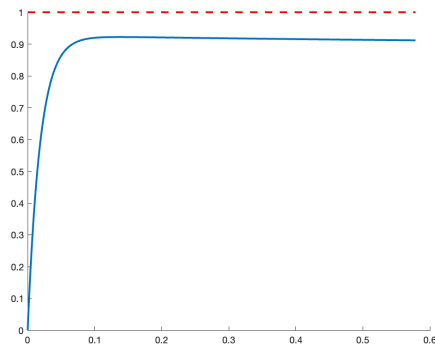
Bonus Problem. (20pt)

(a) (10pt) What are the two criteria for Routh stability condition?

Solution: The first condition is that all the coefficients of the characteristic equation should be positive. Once the first condition is satisfied, we create a Routh table and all the elements in the first column should be positive.

(b) (10pt) The step response of your open loop system is given as follows. What controllers should you use to get rid of the steady state error? (Select all) Explain why.

(1) P (2) PI (3) PD (4) PID (5)None of the four



Solution: (2) and (4). The integral controller can correct the steady state error.